

- 6 *Ensatina eschscholtzii* is a species of salamander that lives in woodland ecosystems in California, USA.

Fig. 6.1 shows an ensatina salamander.

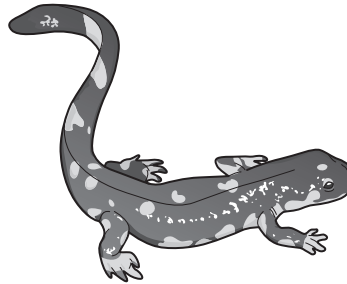


Fig. 6.1

- (a) Define the term ecosystem.

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- (b) The term species can be defined using different concepts, including the biological species concept and the morphological species concept.

The morphological species concept is based on appearance or observable characteristics.

Describe what is meant by the biological species concept.

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- (c) Populations of *E. eschscholtzii* are separated by the Central Valley, shown in Fig. 6.2, which contains conditions unsuitable for salamanders.

Some scientists think that the separated populations of *E. eschscholtzii* have evolved to form different species of salamander.

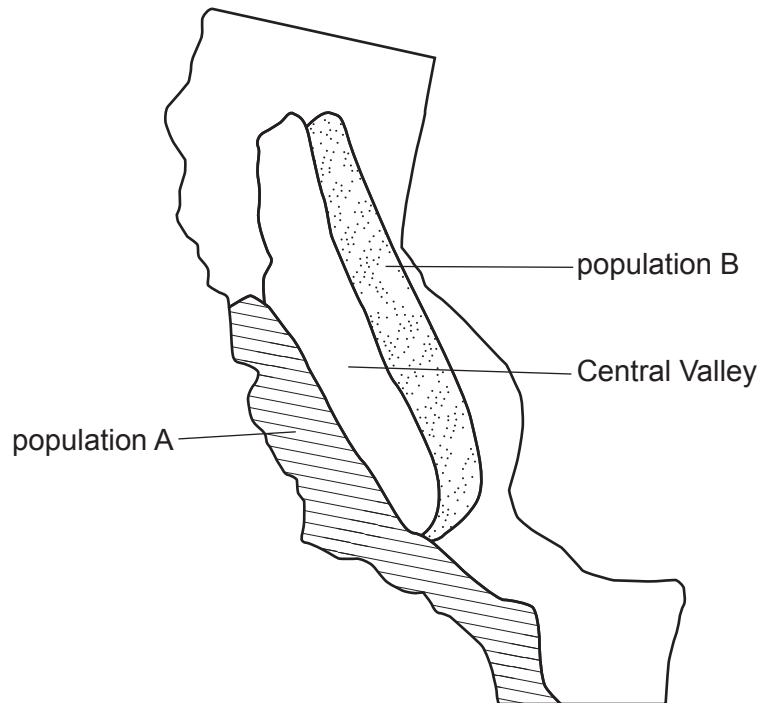


Fig. 6.2

Individuals from different salamander populations may not be able to reproduce with each other, even if they are able to interact.

Explain how salamanders from different populations have evolved to be unable to reproduce with each other.

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